

Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Cómputo en la nube

Actividad 5.8: Instalación de entorno LAMP

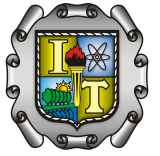
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Introducción

El entorno LAMP es una plataforma de desarrollo web de código abierto ampliamente utilizada en la industria para desplegar aplicaciones y sitios web dinámicos. Sus siglas hacen referencia a los cuatro componentes que lo integran: Linux como sistema operativo, Apache como servidor web, MySQL como gestor de base de datos relacional y PHP como lenguaje de programación del lado del servidor. La combinación de estas tecnologías permite construir una infraestructura funcional, estable y escalable sobre la cual pueden ejecutarse una gran variedad de aplicaciones web. En esta actividad se llevó a cabo la instalación y configuración de dicho entorno sobre una máquina virtual con Ubuntu Server 22.04 LTS alojada en Microsoft Azure, con el objetivo de comprender el proceso de aprovisionamiento de un servidor web en la nube y familiarizarse con las herramientas que conforman este stack tecnológico.

1. Actualización del sistema

```
mauro@CNMV: ~  
mauro@CNMV:~$  
mauro@CNMV:~$ sudo apt update && sudo apt upgrade -y  
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease  
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]  
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]  
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]  
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]  
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]  
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]  
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3440 kB]  
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [519 kB]  
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.7 kB]  
Get:14 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [5826 kB]  
Get:15 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [1102 kB]  
Get:16 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [600 B]  
Get:17 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1268 kB]  
Get:18 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [315 kB]  
Get:19 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [30.5 kB]  
Get:20 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [71.3 kB]  
Get:21 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [15.5 kB]  
Get:22 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [756 B]  
Get:23 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [70.1 kB]  
Get:24 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.4 kB]  
Get:25 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [412 B]  
Get:26 http://azure.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]  
Get:27 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.8 kB]  
Get:28 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.9 kB]
```

2. Instalar Apache

```
Last login: Wed Apr 29 05:28:00 2026 from 189.203.182.90  
mauro@CNMV:~$  
mauro@CNMV:~$ sudo apt install apache2 -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap  
  liblua5.3-0 mailcap mime-support ssl-cert  
Suggested packages:  
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc  
The following NEW packages will be installed:  
  apache2 apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap  
  liblua5.3-0 mailcap mime-support ssl-cert  
0 upgraded, 13 newly installed, 0 to remove and 4 not upgraded.  
Need to get 2153 kB of archives.  
After this operation, 8556 kB of additional disk space will be used.  
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]  
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]  
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]  
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [917 0 B]  
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0 amd64 5.3.6-1build1 [140 kB]  
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.19 [1360 kB]  
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.19 [165 kB]  
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.19 [89.5 kB]  
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all 3.70+nmu1ubuntu1 [23.8 kB]
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
mauro@CNMV:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-04-29 05:34:10 UTC; 1min 20s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 1461 (apache2)
      Tasks: 55 (limit: 1064)
     Memory: 5.2M
        CPU: 52ms
    CGroup: /system.slice/apache2.service
            └─1461 /usr/sbin/apache2 -k start
              └─1463 /usr/sbin/apache2 -k start
                └─1464 /usr/sbin/apache2 -k start

Apr 29 05:34:10 CNMV systemd[1]: Starting The Apache HTTP Server...
Apr 29 05:34:10 CNMV systemd[1]: Started The Apache HTTP Server.
mauro@CNMV:~$
```



3.Instalar php

```
mauro@CNMV:~$ sudo apt install php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.1 php-common php8.1 php8.1-cli php8.1-common php8.1-opcache php8.1-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.1 php php-common php8.1 php8.1-cli php8.1-common php8.1-opcache
  php8.1-readline
0 upgraded, 9 newly installed, 0 to remove and 4 not upgraded.
Need to get 5135 kB of archives.
After this operation, 21.4 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 php-common all 2:92ubuntu1 [12.4 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 php8.1-common amd64 8.1.2-1ubuntu2.23 [1129 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 php8.1-opcache amd64 8.1.2-1ubuntu2.23 [365 kB]
```



```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
mauro@CNMV:~$ php -v
PHP 8.1.2-1ubuntu2.23 (cli) (built: Jan  7 2026 08:37:41) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.1.2, Copyright (c) Zend Technologies
    with Zend OPcache v8.1.2-1ubuntu2.23, Copyright (c), by Zend Technologies
mauro@CNMV:~$
```

Instalar mysql

```
mauro@CNMV:~$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbcgi-fast-perl libbcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl
  libfcgi0ldbl libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite23 libtimedate-perl liburi-perl mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server-8.0
  mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libbusiness-isbn-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libbcgi-fast-perl libbcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl
  libfcgi0ldbl libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite23 libtimedate-perl liburi-perl mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0
  mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 4 not upgraded.
Need to get 29.8 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7212 B]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-core-8.0 amd64 8.0.45-0ubuntu0.22.04
1 [2707 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-8.0 amd64 8.0.45-0ubuntu0.22.04
1 [2707 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server-core-8.0 amd64 8.0.45-0ubuntu0.22.04
1 [2707 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server-8.0 amd64 8.0.45-0ubuntu0.22.04
1 [2707 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server all 8.0.45-0ubuntu0.22.04
1 [2707 kB]
mauro@CNMV:~$
```

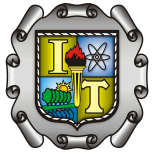
```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
mauro@CNMV:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45-0ubuntu0.22.04.1 (Ubuntu)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```



Conclusión

A través de la realización de esta actividad se logró instalar y verificar el funcionamiento de cada uno de los componentes del entorno LAMP sobre una máquina virtual en Azure. Se comprobó que Apache responde correctamente a peticiones HTTP desde el navegador, que MySQL se encuentra operativo y accesible desde la terminal, y que PHP quedó integrado al servidor web. Este ejercicio permitió reforzar conocimientos sobre administración básica de servidores Linux, manejo de paquetes con apt y gestión de servicios con systemctl. Además, se evidenció la facilidad con la que la nube permite aprovisionar infraestructura de manera rápida y controlada, lo cual representa una ventaja significativa frente a los entornos de servidor tradicionales.